





Management and Curation of EarthPortal Resources

The O'FAIRe properties in a human-readable format

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4 October 2026

1 Introduction

In this short report, our goal is to consider the JSON file that lists the metadata properties read by O'FAIRe [1] and to convert it into a human-readable format. The questions of O'FAIRe name a property in one way and the portal stores it under another, so I need this list at hand to move from one name to the other during my internship.

2 Development

2.1 The file

The file is called `properties.config.json`. I took it from the source code of O'FAIRe [1], as it was on 4 June 2026. It holds 123 entries, one per property. Each entry gives the information of Table 1.

Field	What it says
`portal`	The name of the property in the portal. O'FAIRe keeps the part after the colon and uses it to find the value in the answer of the portal API (for example `omv:hasLicense` gives `hasLicense`).
`mod`	The MOD 1.4 name of the same property. It is the name used in the lists of the questions file.
`type`	`string` for a single value (109 entries) or `array` for several values (14 entries).
`source`	The part of the API answer where the value is read: the submission (99 entries), the ontology record (5), the metrics (10), or the links of the ontology record (9).
`key`	Present in 4 entries only. For `bpm:contact`, it asks for the name of each contact. For three metrics, it is the name under which the portal gives the number (`classes`, `individuals`, `properties`).

Table 1: The fields of one entry. The way O'FAIRe uses them comes from my reading of its Java code.

2.2 The 123 properties

Table 2 gives all the entries, grouped by source and sorted by portal name. I added one column that is not in the file: the questions whose list names the property, taken from the questions file (Report 2).

Table 2: The 123 properties of `properties.config.json`.

Portal name	MOD name	Type	Listed by questions
Read from the submission (99)			
`bpm:authorProperty`	`mod:authorProperty`	string	I3Q3, R1Q1
`bpm:contact`	`dct:rightsHolder`	array (key: `name`)	R1.1Q3
`bpm:creationDate`	`dct:dateSubmitted`	string	F2Q3
`bpm:definitionProperty`	`mod:definitionProperty`	string	I3Q3, R1Q1
`bpm:diffFilePath`	`vann:changes`	string	*none*
`bpm:hierarchyProperty`	`mod:hierarchyProperty`	string	I3Q3, R1Q2
`bpm:notes`	`rdfs:comment`	string	F2Q3
`bpm:notes`	`schema:comment`	string	F2Q3

Portal name	MOD name	Type	Listed by questions
bpm:obsoleteParent	mod:obsoleteParent	string	R1Q2
bpm:obsoleteProperty	mod:obsoleteProperty	string	I3Q3, R1Q1
bpm:prefLabelProperty	mod:prefLabelProperty	string	I3Q3, R1Q1
bpm:publication	dct:bibliographicCitation	string	F2Q3
bpm:pullLocation	omv:resourceLocator	string	F2Q1
bpm:released	dct:created	string	F2Q3
bpm:synonymProperty	mod:synonymProperty	string	I3Q3, R1Q1
cc:morePermissions	cc:morePermissions	string	R1.1Q2
cc:useGuidelines	cc:useGuidelines	string	R1.1Q3
dct:abstract	dct:abstract	string	F2Q3
dct:accrualMethod	dct:accrualMethod	string	R1.2Q3
dct:accrualPeriodicity	dct:accrualPeriodicity	string	R1.2Q3
dct:accrualPolicy	dct:accrualPolicy	string	R1.2Q3
dct:alternative	dct:alternative	string	F2Q1
dct:audience	dct:audience	string	F2Q1
dct:coverage	dct:coverage	string	F2Q1
dct:hasFormat	dct:hasFormat	string	I1Q5, I3Q3
dct:hasPart	dct:hasPart	string	F2Q3, I3Q3
dct:identifier	dct:identifier	array	F1Q2, A1Q1
dct:isFormatOf	dct:isFormatOf	string	I1Q5, I3Q3
dct:publisher	dct:publisher	string	F2Q1
dct:source	dct:source	string	R1.2Q2
dct:valid	dct:valid	string	F2Q3
doap:bugDatabase	doap:bugDatabase	string	F2Q1
doap:mailingList	doap:mailing-list	string	F2Q1
doap:repository	doap:repository	string	F2Q1
door:comesFromTheSameDomain	door:comesFromTheSameDomain	string	F2Q1, I3Q3
door:explanationEvolution	voaf:specializes	string	I2Q6, I3Q3
door:hasDisparateModelling	door:hasDisparateModelling	string	F2Q3, I3Q3
door:isAlignedTo	voaf:hasEquivalencesWith	array	I2Q4, I3Q3
door:ontologyRelatedTo	dct:relation	array	I2Q2, I3Q3
door:similarTo	voaf:similar	array	I2Q6, I3Q3
foaf:depiction	foaf:depiction	string	F2Q3
foaf:fundedBy	foaf:fundedBy	string	R1.2Q8
foaf:homepage	foaf:homepage	string	F2Q1
foaf:logo	foaf:logo	string	F2Q3
idot:exampleIdentifier	idot:exampleIdentifier	string	F2Q1
mod:competencyQuestion	mod:competencyQuestion	string	R1.2Q7
omv:conformsToKnowledgeRepresentationParadigm	omv:conformsToKnowledgeRepresentationParadigm	string	R1.2Q6
omv:description	dct:description	string	F2Q1
omv:designedForOntologyTask	omv:designedForOntologyTask	string	R1.2Q7
omv:documentation	foaf:page	string	F2Q1
omv:endorsedBy	omv:endorsedBy	array	R1.3Q1
omv:hasContributor	dct:contributor	string	R1.2Q1
omv:hasCreator	dct:creator	string	R1.2Q1
omv:hasDomain	dct:subject	string	F2Q1, I3Q3
omv:hasFormalityLevel	omv:hasFormalityLevel	string	I1Q4, I3Q3
omv:hasLicense	dct:license	string	I3Q3, R1.1Q1
omv:hasOntologyLanguage	omv:hasOntologyLanguage	string	I1Q1, I1Q2

Portal name	MOD name	Type	Listed by questions
omv:hasOntologySyntax	omv:hasOntologySyntax	string	I1Q3, I3Q3
omv:hasPriorVersion	owl:priorVersion	string	I3Q3, R1.2Q4
omv:isBackwardCompatibleWith	owl:backwardCompatibleWith	string	F2Q1, I3Q3
omv:isIncompatibleWith	owl:incompatibleWith	array	F2Q3, I3Q3
omv:isOfType	omv:isOfType	string	F2Q3
omv:keyClasses	omv:keyClasses	string	F2Q3
omv:keywords	omv:keywords	string	F2Q1
omv:knownUsage	omv:knownUsage	string	F2Q1
omv:modificationDate	dct:modified	string	F2Q3
omv:naturalLanguage	dct:language	array	F2Q3
omv:status	omv:status	string	A2Q4
omv:URI	owl:ontologyIRI	string	F1Q1, A1Q1, A1Q3
omv:usedOntologyEngineeringMethodology	omv:usedOntologyEngineeringMethodology	string	R1.2Q6
omv:usedOntologyEngineeringTool	omv:usedOntologyEngineeringTool	string	R1.2Q6
omv:useImports	owl:imports	array	I2Q1, I3Q3
omv:version	owl:versionInfo	string	R1.2Q4
owl:deprecated	owl:deprecated	string	A2Q4
owl:versionIRI	owl:versionIRI	string	F1Q4
pav:curatedBy	pav:curatedBy	string	R1.2Q1
pav:curatedOn	pav:curatedOn	string	F2Q3
prov:wasGeneratedBy	prov:wasGeneratedBy	string	R1.2Q2
prov:wasInvalidatedBy	prov:wasInvalidatedBy	string	R1.2Q2
schema:associatedMedia	schema:associatedMedia	string	F2Q3
schema:award	schema:award	string	F2Q3
schema:includedInDataCatalog	schema:includedInDataCatalog	array	F4Q1, F4Q2
schema:translationOfWork	schema:translationOfWork	string	I2Q6, I3Q3
schema:translator	schema:translator	string	R1.2Q1
schema:workTranslation	schema:workTranslation	string	F2Q3, I3Q3
sd:endpoint	sd:endpoint	string	A1Q4
skos:hiddenLabel	skos:hiddenLabel	string	F2Q1
vann:example	vann:example	string	F2Q1
vann:preferredNamespacePrefix	vann:preferredNamespacePrefix	string	F2Q2
vann:preferredNamespaceUri	vann:preferredNamespaceUri	string	F2Q1
voaf:generalizes	voaf:generalizes	string	I2Q6, I3Q3
voaf:hasDisjunctionsWith	voaf:hasDisjunctionsWith	string	F2Q3, I3Q3
voaf:metadataVoc	voaf:metadataVoc	array	I2Q7
voaf:toDoList	voaf:toDoList	string	F2Q3
voaf:usedBy	voaf:usedBy	string	F2Q3, I3Q3
void:dataDump	void:dataDump	string	F2Q3
void:openSearchDescription	void:openSearchDescription	string	F2Q3
void:uriLookupEndpoint	void:uriLookupEndpoint	string	F2Q3
void:uriRegexPattern	void:uriRegexPattern	string	F2Q1
Read from the ontology record (5)			
bpm:group	mod:group	array	R1.3Q2
bpm:viewingRestriction	dct:accessRights	string	R1.1Q2, R1.3Q3
bpm:viewOf	dct:isPartOf	string	I2Q6, I3Q3
omv:acronym	omv:acronym	string	F2Q1
omv:name	dct:title	string	F2Q1

Portal name	MOD name	Type	Listed by questions
Read from the metrics (10)			
bpm:averageChildCount	mod:averageChildCount	string	*none*
bpm:classesWithMoreThan25Children	mod:classesWithMoreThan25Children	string	F2Q3
bpm:classesWithNoDefinition	mod:classesWithNoDefinition	string	F2Q3
bpm:classesWithOneChild	mod:classesWithOneChild	string	F2Q3
bpm:maxChildCount	mod:maxChildCount	string	F2Q3
bpm:maxDepth	mod:maxDepth	string	R1Q2
omv:numberOfAxioms	omv:numberOfAxioms	string	F2Q2
omv:numberOfClasses	omv:numberOfClasses	string (key: classes)	F2Q2
omv:numberOfIndividuals	omv:numberOfIndividuals	string (key: individuals)	F2Q2
omv:numberOfProperties	omv:numberOfProperties	string (key: properties)	F2Q2
Read from the links of the ontology record (9)			
bpm:analytics	mod:analytics	string	F2Q3
bpm:classes	void:classPartition	string	F2Q3
bpm:projects	mod:ontologyInUse	array	R1.3Q1
bpm:properties	void:propertyPartition	string	F2Q3
bpm:reviews	mod:hasEvaluation	array	F2Q1
bpm:roots	void:rootResource	string	F2Q3
bpm:submissions	dct:hasVersion	string	I3Q3, R1.2Q4
bpm:ui	mod:browsingUI	string	F2Q3
mod:metrics	mod:metrics	string	F2Q2

3 Conclusion

Below are some observations that may be useful soon or later in my internship.

- 71 entries have the same name on both sides; the other 52 are stored by the portal under a name that differs from the MOD name.
- The portal name `bpm:notes` appears twice: it is given for `rdfs:comment` and again for `schema:comment`.
- Two MOD names are in no list of the questions file. They are `mod:averageChildCount` and `vann:changes`.
- In the other direction, the list of F2Q3 names `void;classPartition` (with a semicolon), which is not in this file; the file has `void:classPartition`.

References

[1] O'FAIRe configuration files: metadata properties and assessment questions. `https://github.com/agroportal/fairness/tree/master/src/main/resources/config/common`. Files `properties.config.json` and `questions.config.json`, at commit `24def6b` of 4 June 2026.